VoxPulse

Speech Tone & Pace Analyzer

Technical Documentation
Multimodal AI Hackathon 2026

1. Project Overview

VoxPulse is a speech analysis prototype designed to help users understand and improve their speaking delivery. It focuses on common speech-related issues such as speaking too quickly, unnatural pauses, unclear speech, and variations in tone.

The project aims to make speech improvement easier by presenting speech-related information through a simple website dashboard. Users can upload an audio recording or use live recording option, view the dashboard interface, and access an AI chat box for questions about improving their speaking skills.

Project Website: https://vox-pulse-nine.vercel.app/

2. Problem Statement

Effective communication depends not only on the words being spoken but also on the way they are delivered. Speaking too quickly, hesitating frequently, speaking in a monotone voice, or mumbling can make communication less clear and engaging.

VoxPulse aims to provide an accessible way to explore these speech delivery issues and help users understand areas where improvement may be needed.

3. Project Objectives

The main objectives of VoxPulse are:

- To prepare speech samples representing different speaking patterns.
- To explore audio analysis using Python.
- To present speech-related metrics through a user-friendly dashboard.
- To provide an interface for uploading or recording speech.
- To offer an AI chat box where users can ask questions related to voice and speaking improvement.
- To develop a foundation for future integration of actual audio analysis with the website.

4. Website Features

4.1 Audio Upload and Recording

The website provides an option to upload an audio file and an option to record speech live. These features are intended to make speech testing convenient for users.

4.2 Speech Scoring Dashboard

The dashboard displays an overall speech score along with individual metrics related to pace, tone, pauses, and clarity. These scores are currently illustrative demo values used to demonstrate the dashboard interface. They are not generated by Ananya's Python analysis code because the integration has not yet been completed.

4.3 AI Chat Box

VoxPulse includes an AI chat box that users can use to ask questions about speaking and voice improvement. It adds an interactive assistance feature to the website. The exact quality and scope of its responses depend on the implementation available in the current prototype.

4.4 Results Interface

The dashboard provides a visual representation of speech-related metrics. In a future integrated version, the interface can display actual analysis results, detected issues, timestamps, and personalized suggestions generated from the uploaded audio.

5. Audio Dataset Preparation

A small custom dataset of speech samples was prepared for the project. The audio samples were generated using ElevenLabs, an AI voice-generation platform, and organized to represent different speech delivery patterns.

The samples use the same or similar speech content so that different delivery styles can be explored more consistently.

The prepared audio categories are:

Audio sample| Purpose
Normal audio| Baseline example of regular speech delivery
Rushed audio| Example of unusually fast or rushed speaking
Pause and "umm" audio| Example of hesitation and unnatural pauses
Mumbling audio| Example of unclear or less intelligible speech

The samples were organized and their relevant speech issues and timestamps were noted for reference. This preparation supports comparison between normal and flawed speech patterns and can be used for future testing of the analysis system.

The dataset is a preliminary project dataset rather than a large, fully validated benchmark.

6. Python-Based Audio Analysis

Ananya worked on a separate Python-based audio analysis module. The work explores measurable audio characteristics that can help identify speech delivery patterns, including speaking pace, pitch-related variation, volume, and pauses.

The intended role of this module is to process audio and provide information that could be used by the website to display more meaningful speech scores and feedback.

At the current stage, the Python analysis code has not been integrated into the live website. Therefore, the website's illustrative dashboard scores should not be interpreted as results produced by this Python module.

The Python analysis work and prepared audio samples form the basis for further development and testing.

7. System Workflow and Current Integration Status

The project currently has two main parts:

Website prototype: Provides the user interface, audio upload and recording options, illustrative scoring dashboard, and AI chat box.

Python analysis module: Developed separately to explore audio characteristics and support future speech analysis.

The intended future workflow is:

1. The user uploads or records speech.
2. The audio is sent to the Python analysis module.
3. The module processes the audio and identifies relevant speech characteristics.
4. The system returns analysis results, issue timestamps, and possible improvement suggestions.
5. The website displays the results in the dashboard.

This complete workflow is not yet operational end to end, because the website and Python module are currently separate. The existing website is being presented as a prototype, while Python integration remains a planned development step.

8. Team Contributions

Ishita — Dataset Preparation

Prepared and organized AI-generated speech samples using ElevenLabs, categorized the samples by speaking pattern, and noted relevant issues and timestamps for analysis and testing.

Ananya — Python Audio Analysis

Worked on the Python-based audio analysis module to explore speech characteristics and support the future generation of speech-related results.

Anshika — Website Development

Developed the VoxPulse website interface, including the audio upload and recording options, scoring dashboard, and AI chat box.

9. Current Limitations

The current version has the following limitations:

- The Python analysis module is not yet connected to the live website.
- The dashboard scores are illustrative and are not produced by the Python module.
- The prepared dataset is small and requires further testing and validation.
- End-to-end generation of actual speech scores, detected issue timestamps, and personalized feedback through the website remains future work.

These limitations describe the current development stage and help define the next steps for the project.

10. Future Scope

Future development can focus on:

- Integrating the Python analysis module with the website through a backend API.
- Connecting the Analyze Speech action to actual audio processing.
- Generating real scores from measurable audio features.
- Displaying start and end timestamps for detected speech issues.
- Providing explanations and actionable suggestions for each detected issue.
- Testing the system with normal and flawed speech samples.
- Expanding the dataset and evaluating the consistency of analysis results.
- Improving the AI chat box and connecting its guidance more closely with actual analysis results.

11. Conclusion

VoxPulse is a prototype focused on speech delivery analysis and improvement. It combines a user-facing website, an interactive AI chat box, a preliminary dataset of AI-generated speech samples, and separate Python audio analysis work.

The current website demonstrates the intended user experience and scoring interface. The Python module has been developed separately, and its integration with the website remains pending. The prepared audio samples and analysis work provide a starting point for building a more complete system in future development.

The project establishes a foundation for making speech-related feedback more accessible and for further exploring automated speech analysis.